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Concept Review

Lead Through

Why Learn Lead Through?

The Lead Through technique enables providers of robotic systems an ability to program unique tasks to a range of robots. The most common use case corresponds to robotic manipulation where an operator will record a series of waypoints required by the manipulator to perform a specific task. Lead Through applications focus on the manual recording and playback of joint level commands.

Lead Through

Lead Through is used to teach robot tasks such as welding or spray painting, where the path may be customized or non-trivial. The robotic arm is taken through its operating path manually by hand, while the joint position sensors record the joint trajectories. The trajectories are logged at a high enough sample rate in memory such that, when played back, the end-effector will follow a continuous path. This method has the advantage of simplicity but requires reprogramming for each new task. If the operator makes a mistake, the learning process must be restarted.

A Control's Perspective

From a control's perspective, the human operator acts as the controller. The desired position is the operation path that the end-effector must follow. As the operator holds the robotic arm and guides it through the desired position, the human operator essentially applies forces/torques to the manipulator's joints in any direction necessary. The relative position of the end-effector compared to the desired setpoint is captured by human eyes which form a feedback loop to the control system.

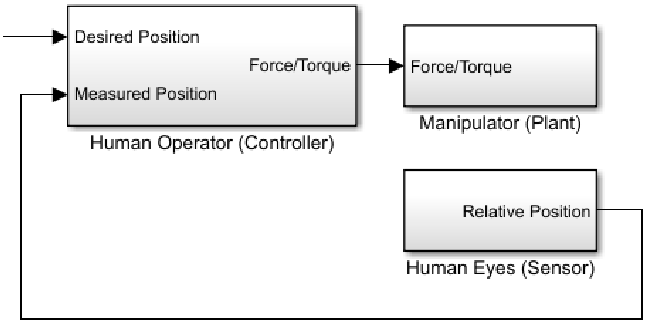

Figure 2 Control Block Diagram for Lead Through

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